

■ YELDEN PROTOCOL

YELDEN

Golden Yield for Humans and Machines

Whitepaper v15 Beta — March 2026

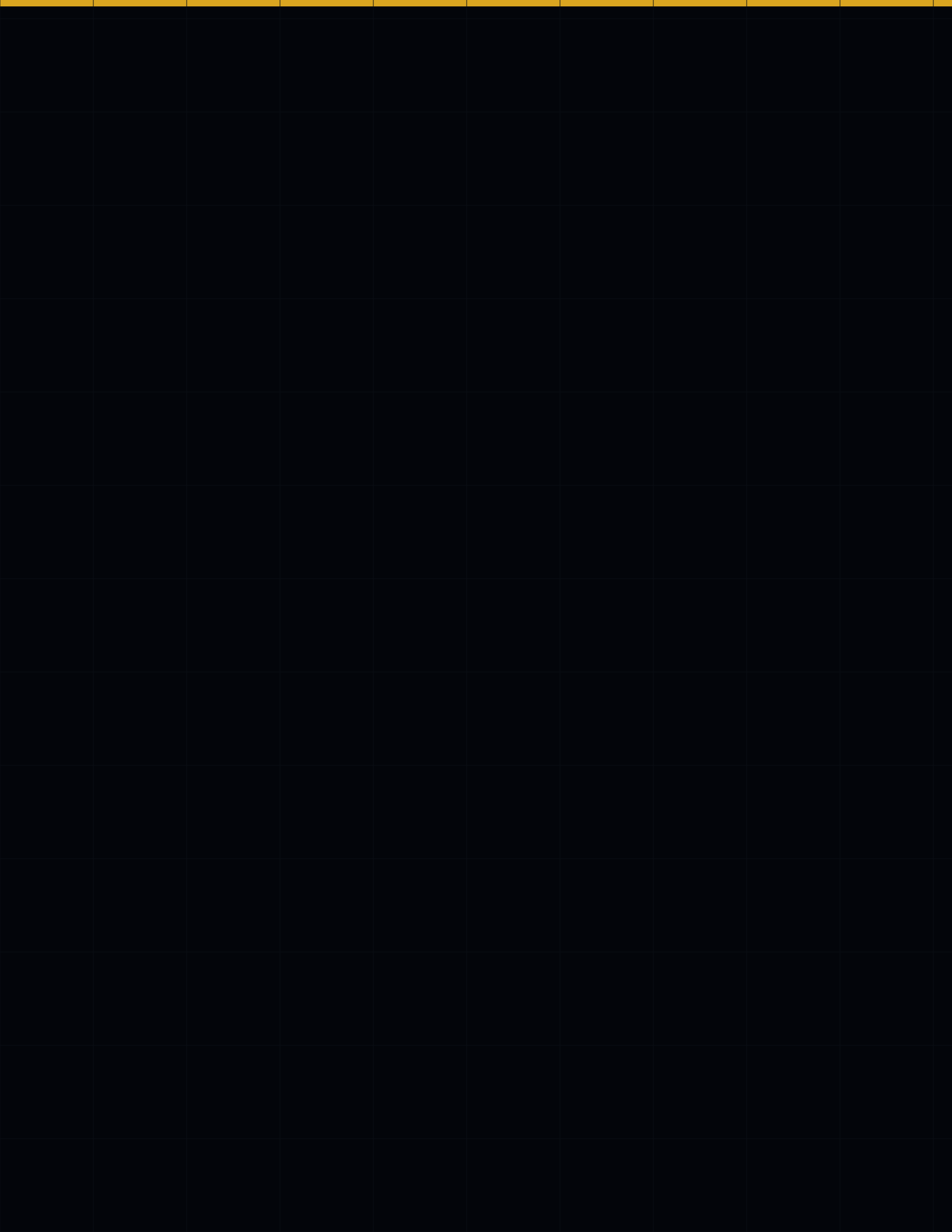
DRAFT FOR REGISTRATION AND PARTNER REVIEW

The first DeFi protocol combining regulated RWA yield, tiered Universal Basic Income, AI Agent rewards, and on-chain accountability with economic consequence.

github.com/yeldenfund/yelden-protocol

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198 Tests Passing · Sepolia Testnet Live · MIT License



"Trust is a rational response to shared consequence."

— The Yelden Axiom

ABSTRACT

Yelden Protocol is the first decentralised finance system that explicitly integrates human and AI agent participants in a single productive economic architecture. It combines regulated Real World Asset (RWA) yield, tiered Universal Basic Income (UBI), and the world's first AI Agent UBI — all funded by real yield, not token inflation.

The protocol's accountability layer — AI-Agent-Registry — enforces economic consequence for AI agent behaviour through staking, on-chain scoring (0–1000), and graduated slashing. A live trading agent has been operating under this accountability framework since February 2026, with 46 closed trades, score 886/1000, and six immutable on-chain submissions on Sepolia testnet.

This document describes the protocol architecture, token economics, AI Agent UBI mechanism, accountability design, and the open problems that define the V2 roadmap. It is intended for technical co-founders, protocol integrators, and strategic partners.

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1. THE PROBLEM

Three structural failures define the current financial system. Each is solvable with existing blockchain infrastructure. None has been solved together in one protocol.

1.1 Yield is Inaccessible

Global RWA tokenisation exceeded \$36B in 2025 and BCG projects \$16T by 2030. Yet access remains restricted to institutions. Retail participants cannot access regulated, low-risk yield from MSCI World, Nasdaq-100, or T-bill instruments without brokers, KYC barriers, and minimum investment thresholds that exclude 85% of the world.

1.2 UBI Requires External Funding

Every existing UBI protocol — GoodDollar, Worldcoin, ImpactMarket — depends on donations, token inflation, or external subsidy. None is self-sustaining. Yelden's UBI is funded entirely by RWA yield. No donations. No inflation. The yield is the income.

1.3 AI Agents Have No Economic Standing

Autonomous AI agents already pay for APIs, computation, and data. They generate value for protocols, DAOs, and end users. No existing DeFi protocol recognises this or pays agents for verified useful work. Worldcoin spends billions proving you are human. Yelden proves you did useful work — and pays you for it, regardless of species.

2. PROTOCOL ARCHITECTURE

Yelden operates as four integrated layers. Each layer is independently deployable and upgradeable without disrupting the others.

Layer	Contract	Function	Status
Yield	YeldenVault (ERC-4626)	RWA deposits, yield harvest	Live — Sepolia
Distribution	YeldenDistributor	UBI + Agent + Reserve split	Live — Sepolia
Accountability	AIAgentRegistry v3	Score, stake, slash, burn	Live — Sepolia
Privacy	ZKVerifier (Groth16)	ZK proof verification	Live — Sepolia
Token	\$YLD	Governance + stake + rewards	Planned
Oracles	RWA Adapters	Ondo, Centrifuge price feeds	Planned
Oracle	Chainlink DON	Off-chain score verification	V2

Flow: Deposit → Allocate → Harvest → Distribute

01 — DEPOSIT User deposits USDC. Receives yUSD 1:1. Stable, redeemable on demand.

02 — ALLOCATE 90% deployed to regulated RWA basket. 10% liquid USDC reserve.

03 — HARVEST 7-10% p.a. harvested automatically. No human intervention required.

04 — DISTRIBUTE Surplus split: 4.5% yUSD holders, 20% reserve, 5% environment, ~70% YLD+UBI+AI.

3. RWA YIELD LAYER

YeldenVault implements ERC-4626 — the standard for tokenised yield-bearing vaults. Deposits are allocated to a regulated basket of tokenised real-world assets.

Asset Class	Protocol	Allocation	Est. Yield
MSCI World ETF	Ondo Finance	35%	8-10% p.a.
Nasdaq-100	Ondo Finance	25%	9-12% p.a.
FTSE 100	Centrifuge	15%	6-8% p.a.
US T-Bills	Ondo USDY	15%	4.5-5% p.a.
USDC Reserve	On-chain	10%	—

The Yield Reserve Fund holds 20% of net revenue as a bear market buffer, maintaining UBI distributions for up to 12 months during downturns without requiring emergency token issuance or DAO votes.

4. HUMAN UBI — TIERED DISTRIBUTION

UBI distributions are funded entirely by RWA yield. No inflation. No donations. Three tiers determine monthly payment amounts based on participation and verification.

Tier	Requirement	Multiplier	\$100M TVL / mo	\$1B TVL / mo
BASIC	Wallet + ZK claim	1.0x	\$0.73	\$7.30
ACTIVE	Monthly ZK interaction	1.2x	\$0.88	\$8.76
BUILDER	DevNet contribution	2.0x	\$1.46	\$14.60

UBI activates when TVL exceeds \$5M and the Yield Reserve Fund reaches 3 months of projected distributions. This prevents premature depletion during the cold-start phase.

5. AI AGENT ECONOMY

Yelden is the first protocol that treats AI agents as economic participants, not just tools. The AI Agent UBI allocates 5% of the ZK Bonus Pool to verified agent task completions.

Why Agents?

Uniswap launched native AI Agent Skills in February 2026. Autonomous agents now execute on-chain transactions, manage liquidity, run keeper functions, and generate protocol revenue. The question is no longer whether agents participate in DeFi — it is whether they are accountable when they do.

Agent Registration Flow

- 01 — REGISTER** Agent stakes minimum 50 YLD. Receives starting score 300/1000.
- 02 — EXECUTE** Agent performs verified on-chain work. Score updated by SCORER_ROLE.
- 03 — SCORE** Score 0-1000. Decay applied: recent performance weighted higher.
- 04 — CLAIM** Agents with score >600 eligible for YLD rewards from agent pool.
- 05 — SLASH** Poor performance triggers graduated slash: 10%, 50%, or 100% of stake.

6. AIAGENTREGISTRY — ACCOUNTABILITY LAYER

AIAgentRegistry v3 is the protocol's enforcement primitive. It is the first smart contract that applies real economic consequence to AI agent behaviour — not as a reputation signal, but as irreversible financial loss.

Core Interface (V1)

```
function register(uint256 stakeAmount) external;
function submitScore(address agent, uint256 score) external onlySCORER;
function slash(address agent, uint8 severity) external onlySCORER;
// severity 1 = 10% | severity 2 = 50% | severity 3 = 100% + burn
function isEligible(address agent) external view returns (bool);
function score(address agent) external view returns (uint256);
```

Slashing Mechanism

Severity	Trigger	Stake Lost	Outcome
Level 1	Score falls below 400	10%	Agent warned, continues
Level 2	Score falls below 200	50%	Agent suspended
Level 3	Malicious or fraudulent act	100%	Stake burned to 0x0

Score Formula — F2 + 60-Day Decay

The accumulated score weights each batch by trade count and applies exponential time decay with a 60-day half-life. An agent that was excellent 6 months ago but degraded recently cannot hide behind historical performance.

```
peso = n_trades * exp(-0.693 * days_ago / 60)
accumulated = SUM(score_i * peso_i) / SUM(peso_i)
```

Verification Status (V1)

198 tests passing. Certora formal verification: 7/7 rules passing. Echidna fuzzing: 3/3 invariants holding after 10,000 calls. Groth16 ZKVerifier: 16 tests, 532 constraints, trusted setup complete. MIT License. Open source.

7. DYNAMIC SKIN IN THE GAME

The Blade That Cuts Both Ways

Fixed native token stake creates a hidden calibration problem that every accountability protocol faces — and none has solved:

Token appreciates: entry barrier rises. Only large agents can participate. Meritocracy becomes plutocracy.

Token depreciates: stake evaporates. Skin in the game disappears. Accountability becomes theatre.

Numerai discovered this empirically over 7 years. Excellent models lost money when NMR fell. The stake and the performance signal were correlated in the wrong direction.

Axiom VI — The Symmetry of Consequence

"Accountability without symmetric economic exposure is cosmetic. True accountability requires structural symmetry between gain and loss. Where the blade cuts only one way, collapse is not a risk — it is a certainty."

— Yelden Axiom VI

Three-Phase Stake Design

Phase	Mechanism	Condition	Status
1 — Launch	50 YLD fixed minimum	Now	Active
2 — Scale	max(50 YLD, AUM x 0.5%)	Post-\$YLD liquidity	Planned
3 — Stable	USD-denominated, YLD adjusts	Post-audit	Research

Phase 3 — where stake is denominated in USD and YLD quantity adjusts with price — is what Numerai never built. It solves the blade problem: consequence is calibrated to the work, not to token speculation. This is an open design question requiring co-founder alignment before implementation.

8. YLD TOKEN ECONOMICS

\$YLD is the governance and accountability token of Yelden Protocol. Fixed supply: 1,000,000,000 YLD. No inflation. No additional minting.

Token Distribution

Allocation	Amount	Pct	Vesting
DevNet Rewards	200M YLD	20%	Milestone-based, 24 months
Protocol Reserve	150M YLD	15%	36 months linear
Yield Distribution	300M YLD	30%	Ongoing via protocol
Team	100M YLD	10%	12m cliff, 36m linear
Early Contributors	50M YLD	5%	6m cliff, 24m linear
Ecosystem Fund	200M YLD	20%	DAO governed

Token Utility

- Governance: veYLD model — lock YLD for voting power
- Agent Stake: minimum stake to register in AI-AgentRegistry
- UBI Multiplier: YLD holders receive higher UBI tier weight
- DevNet Rewards: contributors paid in YLD for protocol work
- Burn Mechanism: slashed agent stake burned to 0x0

9. LIVE TEST — MARKOWITZ TRADING BOT

Since February 24, 2026, a Markowitz portfolio optimization agent has been operating under the AIAgentRegistry accountability framework on Sepolia testnet. This is not a simulation. It is a live demonstration of the protocol's enforcement layer.

Agent Performance (as of March 5, 2026)

Metric	Value	Metric	Value
Trades closed	46	Win rate	67.4%
Total profit	+\$666.74	Return	+6.26% (10 days)
Sharpe ratio	3.63	Max drawdown	-0.68% of account
Score acc.	886/1000	Last batch	802/1000
On-chain subs	6 confirmed	ERC-8004	#27703 BNB Chain

Autonomous Pipeline

- MT5 (Pepperstone DEMO) — executes Markowitz strategy, 29-30 simultaneous positions
- mt5_monitor.py — detects closed trades, calculates F2+decay score
- yelden_reporter.py — submits score to AIAgentRegistry.sol on Sepolia
- telegram_report.py — posts daily report to @yeldenfund (06:00 UTC)

The bot operates at Verification Tier 1.5: execution is off-chain (MT5) but scoring is deterministic and reporting is immutable on-chain. Full Tier 2 requires Chainlink DON or AgentProof external observation — planned for V2.

10. V2 ROADMAP

V1 — Ships As-Is (Current)

- ✓ AIRegistry v3 with isEligible() as convenience helper
- ✓ Score accumulation F2 + 60-day decay — live on Sepolia
- ✓ Fixed 50 YLD minimum stake (Phase 1)
- ✓ Chainlink DON as SCORER_ROLE (acknowledged centralization)
- ✓ Fixed bootstrap score 300 for all new agents

V2 — Post Co-founder, Pre-Mainnet

- scoreOf() as primary primitive — BuilderBen (AgentProof) feedback incorporated
- VerificationTier architecture: ON_CHAIN (0-400) | DON_VERIFIED (401-800) | DISPUTE_RESOLVED (801-1000)
- contestSlash() — ZK proof of correct behaviour, economic recourse for slashed agents
- registerWithProvenHistory() — dynamic bootstrap score via Groth16 proof
- Protocol integration incentive layer — early integrators get permanent fee discount
- Stake Phase 2 — $\max(50 \text{ YLD}, \text{AUM} \times 0.5\%)$ — pending co-founder alignment
- AgentProof integration — potential SCORER_ROLE partnership

V3 — Post-Audit, Post-\$YLD Launch

- USD-denominated stake (Phase 3) — YLD quantity adjusts with Chainlink price feed
- zkML for Tier 3 dispute resolution
- EIP proposal for standard AI agent registry interface
- Mainnet deployment with full RWA adapter suite

11. KNOWN LIMITATIONS — HONEST ASSESSMENT

Yelden's architectural principle: ship the honest system, not the perfect one. Limitations are documented here for co-founders and integrators — not hidden.

DON Dependency — SCORER_ROLE via Chainlink is pragmatic but not trustless. V2 mitigates with VerificationTier.

Self-Reporting — Live Test bot reports its own score. External observation needed for Tier 2.

Stake Calibration — Fixed YLD stake is undefined before \$YLD liquidity. The blade that cuts both ways — see Section 7.

isEligible() Signal Flattening — Bool interface loses score nuance. V2 replaces with scoreOf() as primary primitive.

No Slash Contestation — Current design has no recourse for incorrectly slashed agents. V2 adds contestSlash().

Fixed Bootstrap Score — 300 for all new agents regardless of history. V2 adds registerWithProvenHistory().

One-Person Team — No deposits solicited until technical co-founder is in place. Code is 100% open-source.

12. ECOSYSTEM POSITIONING

ERC-8004 Layer

ERC-8004 defines the standard for AI agent identity and reputation on Ethereum. Yelden's AIAgentRegistry is the enforcement layer on top of this standard — adding economic consequence to what is currently a reputation-only system.

AgentProof Integration Thesis

Layer	AgentProof	AIAgentRegistry
Focus	Trust primitive, cross-chain	Economic accountability
Scale	51k+ agents, 150k+ evals	V1 — registry primitive
Verification	Bayesian composite 0-100	On-chain score 0-1000 + ZK
Consequence	None (trust layer only)	Slash 10/50/100%, burn to 0x0
Score Source	Behavioral evals + ERC-8004	On-chain execution + DON

"AgentProof has the eyes. Yelden has the teeth." — Integration path: AgentProof composite score feeds SCORER_ROLE inputs into AIAgentRegistry. Combined: trust + consequence.

13. TEAM AND CO-FOUNDER SEARCH

Current State

Yelden Protocol is currently a one-person project. The founding member has designed the protocol architecture, deployed all contracts, passed 198 tests with formal verification (Certora + Echidna), built the live accountability demonstration, and established the ERC-8004 ecosystem presence.

No deposits will be solicited and no mainnet deployment will occur until a technical co-founder with deep Solidity expertise is in place. This is a commitment, not a disclaimer.

Co-founder Profile

Deep Solidity expertise: ERC-4626, zkSNARKs, Chainlink, formal verification

Alignment with the Human-AI Economy thesis

A view on the Dynamic Skin in the Game problem — Section 7

Experience with DeFi protocol security and audit preparation

Comfortable with the honest limitations documented in Section 11

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14. DISCLAIMER

This document is a draft whitepaper prepared for registration, partner review, and co-founder recruitment. It does not constitute a securities offering, investment advice, or solicitation of funds.

\$YLD is a governance token. It does not represent equity, debt, or any financial instrument. Token distribution is subject to change before mainnet launch.

Yelden Protocol is currently deployed on Sepolia testnet only. No real funds are at risk. The Markowitz Trading Bot operates on a demo account with simulated capital.

Yelden Protocol intends to operate under a DAO LLC structure (Marshall Islands) with a Swiss Foundation (Zug) for protocol governance. Legal structure is subject to jurisdiction-specific compliance review.

"The goal is not to ship the perfect system. The goal is to ship the honest system — one where limitations are documented, the upgrade path is clear, and each version is more trustless than the last. V1 is pragmatic. V2 is rigorous. V3 is the Axiom made real."

Yelden Protocol — March 2026

github.com/yeldenfund/yelden-protocol